

Isomorphism Theorem.

1st. $G/\ker \phi \cong \phi[G]$ for any Homomorphism $\phi: G \rightarrow G'$,

2nd. If $A, B \leq G$ and $A \leq N_G(B)$, then $AB \leq G$ and $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$, and

$$AB/B \cong A/A \cap B.$$

Proof. $AB = \bigcup_{a \in A} aB = \bigcup_{a \in A} Ba = BA$. Thus $AB \leq G$.

And, $A, B \leq N_G(B)$ implies $AB \leq N_G(B)$, thus $B \trianglelefteq AB$.

$A \cap B \trianglelefteq A$ is clear. Now, a Map

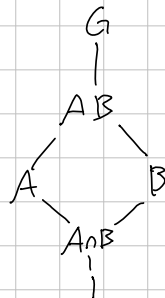
$$\phi: A \rightarrow AB/B: a \mapsto aB$$

is onto Homomorphism with $\ker \phi = A \cap B$.

3rd. If $H, K \leq G$ and $H \leq K$, then $K/H \leq G/H$ and

$$[G/H]/[K/H] \cong G/K.$$

Proof. Take $\phi: G/H \rightarrow G/K: gH \mapsto gK$ as onto Homomorphism with $\ker \phi = K/H$.



4th. Let G be a Group, $N \trianglelefteq G$ be a Normal subgroup.

Denote $\mathcal{S} := \{A \leq G \mid N \leq A \leq G\}$ and $\overline{\mathcal{S}} := \{\overline{A} \leq G/N \mid 1 = N/N \leq \overline{A} \leq G/N\}$.

Then, a Map $\mathcal{Q}: \mathcal{S} \rightarrow \overline{\mathcal{S}}: A \mapsto A/N$ is bijection satisfying: for any $N \leq A, B \leq G$,

$$1) A \leq B \iff \mathcal{Q}(A) \leq \mathcal{Q}(B)$$

$$2) \text{ If } A \leq B, \text{ then } |B:A| = |\mathcal{Q}(B):\mathcal{Q}(A)|.$$

$$3) \mathcal{Q}(\langle A, B \rangle) = \langle \mathcal{Q}(A), \mathcal{Q}(B) \rangle$$

$$4) \mathcal{Q}(A \cap B) = \mathcal{Q}(A) \cap \mathcal{Q}(B)$$

$$5) A \trianglelefteq B \iff \mathcal{Q}(A) \trianglelefteq \mathcal{Q}(B)$$

Proof. Injective: Suppose $\mathcal{Q}(A) = \mathcal{Q}(B) \Leftrightarrow A/N = B/N$.

Let $a \in A$. By assumption, $aN = bN$ for some $b \in B$. Thus, $ab^{-1} \in N$.

Since $N \leq B$, $ab^{-1} \cdot b \in B$. That is, $A \leq B$. Similarly, $A \geq B$. Thus, $A = B$.

Surjective: Let $\overline{A} \in \overline{\mathcal{S}}$ be given. Consider a Map $\pi: G \rightarrow G/N, g \mapsto gN$.

Since $\overline{A} \leq G/N$, the Preimage $\pi^{-1}[\overline{A}]$ is also subgroup of G containing N .

$$\text{Now, } \mathcal{Q}(\pi^{-1}[\overline{A}]) = \pi^{-1}[\overline{A}]/N = \pi^{-1}[\pi^{-1}[\overline{A}]]_{\uparrow N} = \overline{A}$$

Proof 1) $\Rightarrow$ is trivial. Suppose $A/N \leq B/N$. Then, for any $x \in A$, $xN \in B/N$.

That is, $x = bn$ for some $b \in B, n \in N$. Since $N \leq B$, $x = bn \in B$.

Proof 5). Let $A \trianglelefteq B$. For any $xN \in A/N$ and $bN \in B/N$, $bx b^{-1}N \in A/N$.

Let $A/N \trianglelefteq B/N$. For any $x \in A, b \in B$, $bx b^{-1}N \in A/N$, thus $bx b^{-1} \in A$.

Proof 2). Let

$$\psi: \{gA \mid g \in B\} \rightarrow \{gN \in \mathcal{Q}(A) \mid gN \in \mathcal{Q}(B)\}: gA \mapsto gN \in \mathcal{Q}(A).$$

Then, ψ is well-defined injection:

$$g_1N \in \mathcal{Q}(A) = g_2N \in \mathcal{Q}(A) \Leftrightarrow g_1g_2^{-1}N \in \mathcal{Q}(A) = A/N \Leftrightarrow g_1g_2^{-1} \in A \Leftrightarrow g_1A = g_2A$$

And surjection is clear.